EISENSTEIN SOURCES OF THE SPORADIC APÉRY COMPANIONS: WHAT THE SECOND SOLUTION IS MADE OF, AND WHERE ITS LIMIT COMES FROM

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Abstract. For each of the fifteen sporadic Apéry-like recurrences the second ("companion") solution is given by a proved uniform formula $B = F\,\theta_q^{-r}\Phi$ with source $\Phi = t\sigma^r/(PF)$, where $t(q)$ is the nome coordinate and $F(q)$ the parametrizing form. This paper answers the question — posed to the programme by the collaborator Sol — of what modular object Φ actually *is*. Twelve of the fifteen sources are pure Eisenstein series of weight $r+1$, with explicit divisor-sum coefficient laws $c(m)$, and no cuspidal component anywhere [VERIFIED: exact $\mathbb{Q}$, coefficientwise to q^{60}, with the fits found at q^{26} and a held-out band q^{27}–q^{60}]. Cooper's three families admit no such fit in the present coordinate [EXCLUDED: stated bases]. We then state and prove the local *fold connection lemma* $\xi = \Theta(q_c) + F(q_c)\Theta'(q_c)/F'(q_c)$ and verify numerically that all nine known Apéry limits are recovered from the identified source alone (Apéry's $\zeta(3)/6$ to $6.6\cdot 10^{-15}$). The three "no-limit" families are exactly those with a complex-conjugate fold; we record their canonical complex Eichler values to 15, 39 and 33 digits, apparently new constants (PSLQ-negative over natural L-value bases). Structurally: the sporadic class is Eisenstein-sourced, which explains why its limit column consists of zeta and Dirichlet L-values, and which says that a genuinely new elliptic Apéry limit must come from a cuspidal source. Every claim carries its evidence label; the Eisenstein identifications are finite computations, never presented as proofs.

Contents

1. Introduction, for someone who has not met these objects

1.1. Apéry-like recurrences. In 1978 Apéry proved $\zeta(3) \notin \mathbb{Q}$ by exhibiting two solutions of one three-term recurrence,

$$(n+1)^3 A_{n+1} = (2n+1)(17n^2+17n+5)A_n - n^3 A_{n-1}, \tag{1}$$

namely the integer sequence $A_n = \sum_k \binom{n}{k}^2\binom{n+k}{k}^2$ started from $A_0 = 1, A_1 = 5$, and a second solution B_n of the *same* recurrence started from $B_0 = 0, B_1 = 6$, whose entries are rationals

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with controlled denominators. The ratio B_n/A_n converges, very fast, to $\zeta(3)/6$; the speed of the convergence against the size of the denominators is the irrationality proof.

A recurrence of the shape

$$\text{(R2)}\qquad (n+1)^2A_{n+1} = (an^2+an+b)A_n - c\,n^2A_{n-1}, \tag{2}$$

$$\text{(R3)}\qquad (n+1)^3A_{n+1} = (2n+1)(an^2+an+b)A_n - n(cn^2+d)A_{n-1} \tag{3}$$

is called *Apéry-like* when the solution with $A_0 = 1$ is an integer sequence. This is a very restrictive demand. Beukers, Zagier, Almkvist–van Straten and Cooper searched the parameter space; what turned up is a short list of *sporadic* solutions — six of type (R2) (labelled here **A** (Franel), **B**, **C**, **D**, **E**, **F**), six of type (R3) with $d = 0$ (α (Domb), γ (Apéry's own, (1)), δ, ε, ζ, η), and three further (R3) rows with $d \neq 0$ found by Cooper (s_7, s_{10}, s_{18}), which are of *operator order two* despite the cubic-looking recurrence. Fifteen pairs in all. Each one either has a known limit $\lim B_n/A_n$ — always a zeta value or a Dirichlet L-value, e.g. $\zeta(2)/4$, $\zeta(3)/6$, $L(\chi_{-3}, 2)/2$, Catalan's $G/2$ — or is listed in the tables as having "no limit".

Throughout, $r \in \{2, 3\}$ denotes the order of the *operator*, so $r = 2$ for the six (R2) families *and* for Cooper's three, and $r = 3$ for the six order-three families.

1.2. **The companion problem and its solution.** Given a sporadic recurrence, the *companion problem* asks for a closed formula for the second solution B_n. The programme's companion paper [1] solves it uniformly. Write

$$y_0(t) = \sum_{n\geq 0} A_n t^n, \qquad L = \text{the generating operator in } \theta_t = t\,d/dt, \qquad L(y_0) = 0,$$

let $q = t\exp(g/y_0)$ be the *nome* built from the Frobenius companion $y_1 = y_0 \log t + g$, let $t(q)$ be its inverse and $F(q) = y_0(t(q))$ the *parametrizing form*. Let $P(t) = 1 - at + ct^2$ for (R2) and $P(t) = 1 - 2at + ct^2$ for (R3) be the characteristic polynomial appearing as the leading symbol, and let $\sigma = q\,dt/dq \cdot t^{-1}$ (so $\sigma = \theta_q t/t$). Then [1, Thm. 2.4, Thm. 3.3, Thm. 3.5]:

Theorem 1.1 (uniform companion formula; [1])**.** [THM] *For every one of the fifteen sporadic pairs and all $n \geq 0$,*

$$B_n = [t^n]\; F \cdot \theta_q^{-r}\Phi, \qquad \boxed{\Phi = \frac{t\,\sigma^r}{P(t)\,F}}$$

where θ_q^{-1} is the constant-free antiderivative $\sum b_m q^m \mapsto \sum (b_m/m) q^m$.

We do not reprove this here; the proof is two lemmas (a boundary-defect computation $L(y_B) = t$, and a uniqueness statement for log-free solutions) plus the fact that all fifteen operators are *exactly rectified*, $L = C(t)F\theta_q^r F^{-1}$, which for $r = 3$ is the symmetric-square theorem of [1]. The reader should simply take Theorem 1.1 as given: *the companion is an r-fold q-antiderivative of one q-series Φ, times F.* In the classical language, B is an *Eichler integral* of Φ.

1.3. **The question of this paper.** Theorem 1.1 turns the companion problem into a question about a single generating function. The source Φ is manufactured from t, F and P; by the general Beukers picture one expects it to be modular of weight $r + 1$ on the level of the family. Sol's question, which organizes this paper, is the sharp version:

> *Which modular form is Φ? Given its coefficients, can the Apéry limit be read off as modular-transformation data of the source?*

The answers, in order: (i) twelve of fifteen sources are *pure Eisenstein* of weight $r + 1$, with a closed divisor-sum coefficient law (§3); (ii) yes — the limit is the value at the fold of the Eichler integral, by an explicit connection formula (§4), and this reproduces all nine known limits from the source alone; (iii) the three families with "no limit" are exactly the three whose fold is at a complex point, and their connection values are computable, apparently new constants (§5).

1.4. **Evidence discipline.** This programme labels every statement. [THM] and [PROVED] mean a proof exists. [VERIFIED: ...] means an exact but finite computation over the stated range; **a finite verification is never stated as a proof**, and prose about it never becomes stronger than the label. [EXCLUDED: ...] is a proved (or exhaustively searched) negative within the stated search space. [OPEN] is open. Section 8 lists every claim of this paper with its label; §7 states precisely what is missing between the verifications and theorems.

The identifications below are exact rational computations (`Fraction` arithmetic throughout, no floating point) carried to q^{60}, with the fits determined by the coefficients up to q^{26} and the remaining 34 coefficients held out as a blind test. That is strong evidence and not a proof. What separates it from a proof is spelled out in §7, and it is not much: Sturm bounds in the relevant spaces are all ≤ 18.

1.5. **Credit.** The programme of identifying Φ as a modular object and deriving the limits from it was proposed by the collaborator Sol; steps 1, 2 and 4 of that proposal are what this paper executes. The base theorem is [1]. The (t, F) identifications used as coordinates are from the programme's modular dictionary [2]. Source scripts and ledger: `work/PHI_SOURCE_LEDGER.md`, `work/z5eps/eps60_phi_source.py`, `eps60b_phi_holdouts.py`, `eps61_eichler_limits.py`, `eps61b_complex_l`

2. Coordinates: the nome, the form, and the source

Everything below happens in the q-coordinate. Two facts from [2] set the stage.

Computer-assisted finite verification 2.1 (the dictionary)**.** [VERIFIED: exact $\mathbb{Q}$, q^{25}, zero-tail-checked; [2]] All fifteen nomes are integral at scale $\mu = 1$ (that is, $t(q) \in q\mathbb{Z}[[q]]$ with no rescaling $q \mapsto \mu q$ needed), and twelve of the fifteen pairs (t, F) are exactly identified as (generalized) eta quotients on an explicit level, F of weight 1 for the (R2) six and weight 2 for the order-three six. For instance

$$\gamma: \quad t = q\Big(\frac{\eta_1\eta_6}{\eta_2\eta_3}\Big)^{12}, \qquad F = \frac{(\eta_2\eta_3)^7}{(\eta_1\eta_6)^5} \quad \text{(level 6)},$$

which is Beukers' classical parametrization of Apéry's $\zeta(3)$ case, here re-derived from the recurrence. Cooper's three (s_7, s_{10}, s_{18}) have integral but *aperiodic* t and F in this coordinate: not eta quotients as normalized here.

Since F has weight 1 (resp. 2) and $\sigma = \theta_q t/t$ has weight 2 as a logarithmic derivative of a hauptmodul, the shape $\Phi = t\sigma^r/(PF)$ has, formally, weight $2r - \operatorname{wt} F = r + 1$ in both cases: weight 3 for the (R2) six, weight 4 for the order-three six. That is the prediction. What is not predicted, and is the content of §3, is that Φ lies in the *Eisenstein* part.

Remark 2.2. Note the normalization $\Phi = q + O(q^2)$: the constant term is 0 and the q-coefficient is 1, forced by $t = q + O(q^2)$, $F = 1 + O(q)$, $P = 1 + O(t)$. Any identification of Φ as a linear combination of Eisenstein series must therefore have its constant terms cancel identically. They do, in every family — a nontrivial internal check, since the constant terms are not part of the fitted data.

2.1. **Divisor sums, and the two positions of the character.** We fix notation once. For a Dirichlet character χ and $k \geq 0$,

$$\sigma_\chi^{(k)}(m) = \sum_{d|m} \chi(m/d)\, d^k \quad \text{(“inner” or mode 2)}, \qquad \widetilde{\sigma}_\chi^{(k)}(m) = \sum_{d|m} \chi(d)\, d^k \quad \text{(“outer” or mode 1)},$$

and $\sigma_3(m) = \sum_{d|m} d^3$ for the trivial character. These are the q-expansion coefficients of the standard Eisenstein series $E_{k+1}^{\chi,\psi}(q) = \text{const} + \sum_m \big(\sum_{d|m} \psi(d)\chi(m/d)d^k\big)q^m$ of weight $k+1$, nebentypus

TABLE 1. The twelve identified companion sources $\Phi = \sum_{m\geq 1} c(m)q^m$. Weight is $r+1$; $\sigma(m/k) := 0$ unless $k \mid m$. [VERIFIED: q^{60}, held-out q^{27}–q^{60}]

fam	r	wt	$c(m)$	level
A	2	3	$\widetilde{\sigma}^{(2)}_{\chi_{-3}}(m) - \widetilde{\sigma}^{(2)}_{\chi_{-3}}(m/2)$	6
B	2	3	$\sigma^{(2)}_{\chi_{-3}}(m) - 6\sigma^{(2)}_{\chi_{-3}}(m/2) - 8\sigma^{(2)}_{\chi_{-3}}(m/4)$	36
C	2	3	$\sigma^{(2)}_{\chi_{-3}}(m) - 8\sigma^{(2)}_{\chi_{-3}}(m/2)$	6
D	2	3	$\sum_{d\mid m}\big(\psi_1(d) - 2\psi_2(d)\big)d^2, \quad \psi_1,\psi_2 = \mathrm{Re}, \mathrm{Im}$ of the odd character mod 5 with $\chi(2) = i$	5
E	2	3	$\sigma^{(2)}_{\chi_{-4}}(m) - 8\sigma^{(2)}_{\chi_{-4}}(m/2)$	8
F	2	3	$\sigma^{(2)}_{\chi_{-3}}(m) - 7\sigma^{(2)}_{\chi_{-3}}(m/2) - 8\sigma^{(2)}_{\chi_{-3}}(m/4)$	12
α	3	4	$\sigma_3(m) - 17\sigma_3(m/2) - 9\sigma_3(m/3) + 16\sigma_3(m/4) + 153\sigma_3(m/6) - 144\sigma_3(m/12)$	12
γ	3	4	$\sigma_3(m) - 28\sigma_3(m/2) + 63\sigma_3(m/3) - 36\sigma_3(m/6)$	6
δ	3	4	$\sigma_3(m) - 14\sigma_3(m/2) - \sigma_3(m/3) + 16\sigma_3(m/4) + 14\sigma_3(m/6) - 16\sigma_3(m/12)$	12
ε	3	4	$\sigma_3(m) - 21\sigma_3(m/2) + 84\sigma_3(m/4) - 64\sigma_3(m/8)$	8
ζ	3	4	$\sum_{d\mid m} \chi_{-3}(d)\,\chi_{-3}(m/d)\,d^3$ (coefficient exactly 1)	9
η	3	4	$\sigma^{(3)}_{\chi_5}(m) - 14\sigma^{(3)}_{\chi_5}(m/2) - 16\sigma^{(3)}_{\chi_5}(m/4)$	20

$\chi\psi$; a level-N Eisenstein space is spanned by such series together with their $q \mapsto q^\ell$ scalings, $\ell \mid N$. We write $\sigma(m/k) := 0$ unless $k \mid m$, so that "$\sigma_3(m/2)$" *is* the coefficient of the scaled series $E_4(q^2)$.

Two characters need names: χ_{-3}, χ_{-4} are the quadratic characters of conductor $3, 4$; χ_5 the quadratic character mod 5; and ψ_1, ψ_2 are the real and imaginary parts of the odd character mod 5 normalized by $\chi(2) = i$, so ψ_1 is supported on ± 1 and ψ_2 on ± 2 modulo 5.

3. The main result: twelve Eisenstein sources

3.1. The table.

Computer-assisted finite verification 3.1 (Eisenstein identification of the sources)**.** [VERIFIED: exact $\mathbb{Q}$, coefficientwise to q^{60}; fits determined at q^{26}, held-out band q^{27}–q^{60} (34 coefficients); `eps60`, `eps60b`] For each of the twelve families listed in Table 1, the companion source $\Phi = \sum_{m\geq 1} c(m)q^m$ of Theorem 1.1 has the stated divisor-sum coefficients $c(m)$, exactly, for every $m \leq 60$; equivalently Φ is the displayed rational combination of weight-$(r+1)$ Eisenstein series on the stated level, **with no cuspidal component**. No rescaling is needed ($\mu = 1$): Φ is already integral in the canonical nome.

Some orientation on Table 1, for a reader meeting it cold.

- **Read a row as a form.** The row for γ says

$$\Phi_\gamma = \tfrac{1}{240}\big(-E_4(q) + 28E_4(q^2) - 63E_4(q^3) + 36E_4(q^6)\big)$$

up to the normalization that makes the constant terms cancel and $\Phi = q + \cdots$; the combination has $240\cdot$ the raw fitted coefficients, and the constant terms cancel identically, as they must.

This is exactly Beukers' weight-4 level-6 Eisenstein series for Apéry's $\zeta(3)$. That we arrive at it *from the recurrence alone*, with no classical input, is the control that validates the method.

- **D likewise reproduces** the $\Gamma_1(5)$ weight-3 picture for $\zeta(2)$ — the second classical control.
- **ζ is the cleanest object in the table:** the primitive (χ_{-3}, χ_{-3})-Eisenstein series of weight 4 and level 9, with coefficient exactly 1 and nothing else. A single Eisenstein newform, on the nose.
- **Nothing cuspidal is ever used.** The fitting bases offered to the search included cusp forms at each level: $\eta_1^4\eta_5^4$ (level 5, weight 4), $(\eta_1\eta_2\eta_3\eta_6)^2$ (level 6), $(\eta_2\eta_4)^4$ (level 8), η_3^8 (level 9), and their $q \mapsto q^\ell$ embeddings. *No fit ever took a cusp form.* The cuspidal coefficient came out 0 in every family, exactly, at every order.

3.2. **Why "held-out" matters.** The fits are linear algebra over $\mathbb{Q}$: a basis of the weight-$(r+1)$ space on the level, and Φ's first 26 coefficients. With bases of dimension ≤ 6, a fit through 26 coefficients is already overdetermined by a lot. But the programme's discipline is to state the evidence as it was actually obtained, so the fits were frozen at q^{26} and then tested against the 34 coefficients $q^{27}, \ldots, q^{60}$, computed afterwards and not used in the fit. All twelve pass with zero residual. This is the same protocol used for the programme's prime tests.

3.3. **Cooper's three.**

Computer-assisted finite verification 3.2 (Cooper's three do not fit here)**.** [EXCLUDED: stated bases; [OPEN] in general] For s_7, s_{10}, s_{18} no fit exists at weight 3 or weight 4 in the Eisenstein spaces of levels $\{7, 10, 18\}$ and their divisors with the natural characters (χ_{-7}, the mod-5 pair, χ_{-3}).

These are precisely the three families whose t and F were already aperiodic in the canonical nome (Verification 2.1). Their Φ *is* integral ($\mu = 1$), so the obstruction is not integrality: it is the *coordinate*. What is presumably needed is an Atkin–Lehner normalization or a different hauptmodul — the same missing ingredient as in the dictionary. It would be wrong to say "Cooper's three are not Eisenstein"; the honest statement is the one labelled above.

Problem 3.3 (Φ-2)**.** [OPEN] Find the coordinate (hauptmodul / Atkin–Lehner normalization) in which t_{s_7}, F_{s_7} are eta-type, and identify Φ_{s_7} there. Same for s_{10}, s_{18}.

4. From source to limit: the fold connection formula

We now do Sol's step 4: read the Apéry limit off the source.

4.1. **The geometry.** Let t_c be the singularity of the family, i.e. the root of $P(t)$ of smallest modulus. The generating functions $y_0(t) = \sum A_n t^n$ and $y_B(t) = \sum B_n t^n$ have radius of convergence $|t_c|$, and

$$\lim_{n\to\infty} \frac{B_n}{A_n}$$

exists (when t_c is real positive) and equals the ratio of the leading *singular* parts of y_B and y_0 at t_c, because the coefficient asymptotics of a function are governed by its dominant singularity, and the non-singular parts contribute exponentially less.

The key structural fact is that in the q-coordinate the map $q \mapsto t(q)$ *folds* at $q_c :=$ the nome of t_c: one has $dt/dq = 0$ there, and $t - t_c \simeq \kappa\,(q - q_c)^2$ with $\kappa \neq 0$ — a simple fold. This is the modular reason a square-root singularity appears in t: the two sheets of $\sqrt{t - t_c}$ are the two branches of q around the fold, exchanged by the local involution.

Lemma 4.1 (fold connection lemma)**.** [PROVED] *Let $y_0(t), y_B(t)$ be the two solutions above, let $t(q)$ be analytic near q_c with a simple fold at q_c, and suppose $y_0(t(q))$ and $y_B(t(q))$ are analytic at q_c (which they are: $y_0(t(q)) = F(q)$ and $y_B(t(q)) = F\theta_q^{-r}\Phi$ by Theorem 1.1, both q-analytic wherever Φ is). Then the singular parts of y_0, y_B at t_c are proportional to the* odd *parts of F, $F\theta_q^{-r}\Phi$ under the local fold involution, and consequently*

$$\xi := \lim_{n\to\infty} \frac{B_n}{A_n} = \frac{\frac{d}{dq}\big(y_B \circ t\big)(q_c)}{\frac{d}{dq}\big(y_0 \circ t\big)(q_c)} = \Theta(q_c) + \frac{F(q_c)\,\Theta'(q_c)}{F'(q_c)}, \qquad \Theta := \theta_q^{-r}\Phi = \sum_{m\geq 1} \frac{c(m)}{m^r} q^m.$$

Proof. Work locally at q_c; write $u = q - q_c$. Since the fold is simple, the local involution ι with $t(\iota q) = t(q)$ is an analytic involution fixing q_c with $\iota(u) = -u + O(u^2)$; after an analytic change of coordinate $u \mapsto v$ (Koenigs' linearization of an analytic involution, elementary in this setting) we may take $\iota(v) = -v$ and $t - t_c = \kappa v^2$ with $\kappa \neq 0$. Then $s := \sqrt{(t - t_c)/\kappa} = v$ is a local coordinate on each sheet, and a function h analytic at q_c decomposes uniquely as

$$h = h^{\mathrm{ev}} + h^{\mathrm{odd}}, \qquad h^{\mathrm{ev}}(v) = \tfrac{1}{2}(h(v) + h(-v)), \quad h^{\mathrm{odd}}(v) = \tfrac{1}{2}(h(v) - h(-v)).$$

The even part is an analytic function of v^2, hence of t: it is the part of h that is regular at t_c. The odd part is v times an analytic function of v^2, i.e. $\sqrt{t - t_c}$ times an analytic function of t: it is exactly the square-root singular part. Since $y_0 \circ t$ and $y_B \circ t$ are analytic in q at q_c, both decompose this way, and their singular parts at t_c are their odd parts.

Now the coefficient asymptotics. If $y(t) = g(t) + \sqrt{1 - t/t_c}\, h(t)$ near t_c with g, h analytic, then by singularity analysis $[t^n]y \sim -h(t_c)\, t_c^{-n} n^{-3/2}/(2\sqrt{\pi})$ (times $1 + O(1/n)$), provided $h(t_c) \neq 0$; the analytic part g contributes $O(\rho^{-n})$ with $\rho > |t_c|$. The same $n^{-3/2} t_c^{-n}$ prefactor occurs for y_0 and y_B, hence

$$\xi = \lim \frac{B_n}{A_n} = \frac{h_B(t_c)}{h_0(t_c)} = \frac{(y_B \circ t)^{\mathrm{odd}\prime}(q_c)}{(y_0 \circ t)^{\mathrm{odd}\prime}(q_c)} = \frac{(y_B \circ t)'(q_c)}{(y_0 \circ t)'(q_c)},$$

the last step because the even parts have vanishing v-derivative at $v = 0$, so the full q-derivative at q_c equals the odd part's; and dv/dq is a nonzero constant at q_c, cancelling in the ratio.

Finally substitute $y_0 \circ t = F$ and $y_B \circ t = F\Theta$ from Theorem 1.1:

$$\xi = \frac{(F\Theta)'(q_c)}{F'(q_c)} = \frac{F'(q_c)\Theta(q_c) + F(q_c)\Theta'(q_c)}{F'(q_c)} = \Theta(q_c) + \frac{F(q_c)\Theta'(q_c)}{F'(q_c)}. \qquad \square$$

Remark 4.2. Two hypotheses are doing work and should be flagged. First, $h_0(t_c) \neq 0$ — i.e. F genuinely has an odd part, equivalently $F'(q_c) \neq 0$; this is checked numerically in every family and is visible from the eta formulas. Second, Θ must extend analytically to q_c; for $|q_c| < 1$ this is automatic from convergence of $\sum c(m) m^{-r} q^m$, since $c(m) = O(m^{r+\epsilon})$ makes the series converge absolutely on $|q| < 1$. Both are satisfied in every case below. The lemma is the "ξ is a ratio of q-derivatives at the fold" statement recorded in the programme's ledgers; this is its written-out proof.

Remark 4.3 (what the formula *is*). $\Theta = \theta_q^{-r}\Phi$ is the Eichler integral of the weight-$(r+1)$ Eisenstein series Φ: its $(r-1)$-fold antiderivative in the classical normalization, $\sum c(m) m^{-r} q^m$. Evaluating an Eichler integral of an Eisenstein series at a CM-like or cuspidal point is exactly the operation that produces L-values of the associated characters — $\sum \sigma_k(m) m^{-r} q^m$ at $q \to$ a cusp degenerates into $\zeta(r)\zeta(r-k)$-type products. That is the conceptual answer to "why are the sporadic limits zeta and Dirichlet L-values": *because the sources are Eisenstein.* See §6.

4.2. **The numerical verification.** With $c(m)$ known in closed form for *all* m — which is the whole point of Table 1, since it makes $\Theta(q_c)$ a computable number rather than a truncated experiment — and with q_c solved by Newton's method from the exact 26-term series $t(q)$ (so the error in q_c is tracked as $O(|q_c|^{26})$), Lemma 4.1 can be evaluated.

TABLE 2. The Apéry limits recomputed from the identified Eisenstein source alone. Starred rows have larger $|q_c|$, where the discrepancy is consistent with the $|q_c|^{26}$ truncation of the series for q_c and F; the four sharp rows are the strong test. [VERIFIED: numeric, tracked truncation]

fam	known limit	$\lvert\xi(\text{from }\Phi) - \text{known}\rvert$
γ	$\zeta(3)/6$	$6.6 \cdot 10^{-15}$
ε	$7\zeta(3)/32$	$8.8 \cdot 10^{-13}$
α	$7\zeta(3)/24$	$8.3 \cdot 10^{-8}$
D	$\zeta(2)/5$	$3.7 \cdot 10^{-7}$
ζ	$L(\chi_{-3}, 3)/3$	$4.7 \cdot 10^{-4}$ *
A	$\zeta(2)/4$	$4.4 \cdot 10^{-3}$ *
C	$L(\chi_{-3}, 2)/2$	$4.4 \cdot 10^{-3}$ *
E	$G/2$ (Catalan)	$5.8 \cdot 10^{-3}$ *
F	$5L(\chi_{-3}, 2)/8$	$3.6 \cdot 10^{-2}$ *

Computer-assisted finite verification 4.4 (all nine limits from the source)**.** [VERIFIED: numeric, series order $N = 26$, error tracked as $|q_c|^N$; eps61] For each of the nine families with a known real limit, the value ξ computed from Table 1 and Lemma 4.1 agrees with the known limit to the accuracy in Table 2.

The right column is not a measure of the theory's accuracy; it is a measure of how far $|q_c|$ is from 0 at series order 26. For γ, where $t_c = 17 - 12\sqrt{2} \approx 0.0294$ is tiny, the agreement is 15 digits and is a genuinely sharp test. For **F**, $|q_c|$ is large enough that $|q_c|^{26}$ is itself of order 10^{-2}, and the row confirms the mechanism rather than the digits. The honest summary is the one stated in the verification label: *all nine limit families reproduce their limits from the identified Eisenstein source*, at the precision each family's geometry allows at this series order. Sol's step 4 holds across the board.

5. The three complex folds: B, δ, η

5.1. **Why there is "no limit".** Three sporadic families are listed in the classical tables as having no limit: **B**, δ, η. Table 1 identifies their sources perfectly well, so nothing has failed; the explanation is elementary and geometric.

Observation 5.1. [PROVED] (immediate from the parameters) $\mathbf{B}, \delta, \eta$ are exactly the sporadic families whose $P(t)$ has a complex-conjugate pair of roots:

$$\mathbf{B}: \ P = 1 - 9t + 27t^2, \ \text{disc} = -27; \qquad \delta: \ P = 1 - 14t + 81t^2, \ \text{disc} = -128; \qquad \eta: \ P = 1 - 22t + 125t^2, \ \text{disc} = -16,$$

with splitting fields $\mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-2}), \mathbb{Q}(i)$ respectively.

With two dominant singularities of equal modulus and non-real ratio, the coefficient asymptotics of y_B and y_0 acquire an oscillating phase and B_n/A_n does not converge — it spirals. That is all "no limit" means. The *connection value* at either singularity is perfectly well-defined: Lemma 4.1's formula $\xi = \Theta(q_c) + F\Theta'/F'$ makes sense verbatim with $q_c \in \mathbb{C}$, and gives a canonical complex number attached to the family (the two roots give complex-conjugate values; we take the one with $\operatorname{Im} t_c < 0$).

Computer-assisted finite verification 5.2 (complex Eichler values)**.** [VERIFIED: numeric, series order $N = 60$, `mpmath` dps 80, errors as stated; `eps61b`] With the sources of Table 1,

$$\begin{aligned}
\xi_\delta &= 0.2893840692791852173287219466486021576717328873 \\
&\quad - 0.3827935392630498091835715898605831841400945 33\,i \qquad (\text{err} \le 3 \cdot 10^{-39}), \\
\xi_\eta &= 0.4274123841779312131535688439535399101269635 19 \\
&\quad - 0.2218628557422189242376759231839914020522177 49\,i \qquad (\text{err} \le 2 \cdot 10^{-33}), \\
\xi_{\mathbf{B}} &= 0.39151062416206523 - 0.42193463325508115\,i \qquad (\text{err} \le 8 \cdot 10^{-15}).
\end{aligned}$$

(**B** is the hard one: $|q_c| \approx 0.58$, so the series truncation at $N = 60$ costs most of the precision. δ and η are good to 39 and 33 digits respectively.)

5.2. These appear to be new constants.

Computer-assisted finite verification 5.3 (PSLQ negative)**.** [EXCLUDED: PSLQ over the stated bases, max coefficient 10^6–10^7] No rational relation was found for $\operatorname{Re}\xi, \operatorname{Im}\xi$ (nor for the CM-twisted variants $\sqrt{2}\operatorname{Re}\xi_\delta$, $\sqrt{5}\operatorname{Re}\xi_\eta$, etc.) over the basis

$$\{1,\ \zeta(3),\ \pi^3,\ \pi^3/\sqrt{2},\ \pi^3/\sqrt{5},\ L(\chi_{-8}, 3),\ L(\chi_{-4}, 3),\ L(\chi_5, 3),\ \pi^2 \log 2, \dots\}.$$

All apparent hits were degenerate. (Caution for anyone repeating this: $L(\chi_{-4}, 3) = \pi^3/32$, so it must not appear alongside π^3 in a basis, or PSLQ returns spurious relations.)

So, to current knowledge, these are new constants: *Eichler values of explicit Eisenstein series at complex-multiplication-adjacent points.* Their plausible home is not the Dirichlet L-value world but periods of Hecke characters — Chowla–Selberg / CM Γ-value territory, where $\mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-2}), \mathbb{Q}(i)$ contribute Γ-quotients rather than $L(\chi, s)$. That is a guess, not a result. The sharpest open form:

Problem 5.4 (Φ-1)**.** [OPEN] Recognize ξ_δ (39 digits in Verification 5.2) in closed form. The source is the explicit level-12 weight-4 Eisenstein combination $\sigma_3(m) - 14\sigma_3(m/2) - \sigma_3(m/3) + 16\sigma_3(m/4) + 14\sigma_3(m/6) - 16\sigma_3(m/12)$ of Table 1; q_c is the nome of the fold over $t_c = (7 - \sqrt{-32})/81$.

δ is the right target of the three: the most digits, the cleanest field ($\mathbb{Q}(\sqrt{-2})$), trivial character, and an identified level-12 eta parametrization for both t and F.

6. Structural consequences

6.1. The sporadic class is Eisenstein-sourced. Combining Theorem 1.1 with Verification 3.1, the whole chain

$$\text{recurrence} \ \longrightarrow\ \text{operator } L \ \longrightarrow\ (t, F) \ \longrightarrow\ \Phi \ \longrightarrow\ \xi$$

is now explicit for twelve of the fifteen families, and *Eisenstein at every link*: F is an Eisenstein eta quotient of weight 1 or 2, t is a hauptmodul, Φ is a pure Eisenstein series of weight $r + 1$, and ξ is the value of its Eichler integral at the fold. The sporadic table is not fifteen coincidences; it is one mechanism run on twelve levels.

6.2. Why the limits are zeta and Dirichlet L-values. This is now a structural statement rather than an observation about a table. $\Theta = \sum c(m) m^{-r} q^m$ with $c(m)$ a divisor sum $\sum_{d|m} \chi_1(d)\chi_2(m/d) d^k$, $k = r - 1$. Formally at $q \to 1$ (that is, at the cusp, which is where the fold sits after the modular transformation to the local coordinate),

$$\sum_m \Big(\sum_{d|m} \chi_1(d)\chi_2(m/d) d^k \Big) m^{-r} q^m \ \longrightarrow\ \text{a product } L(\chi_1, r - k)\, L(\chi_2, r)$$

by Dirichlet-series factorization of the divisor convolution. The Eichler integral of an Eisenstein series has, by the classical Eichler–Shimura mechanism, a period polynomial whose coefficients are exactly critical L-values of the constituent characters; the fold value is a specialization of that data. Hence $\zeta(3)$ for the trivial-character weight-4 families $(\gamma, \alpha, \varepsilon)$, $\zeta(2)$ for the trivial-character weight-3 ones $(\mathbf{A}, \mathbf{D})$, $L(\chi_{-3}, 2)$ and $L(\chi_{-3}, 3)$ for the χ_{-3}-twisted ones $(\mathbf{C}, \mathbf{F}, \zeta)$, and Catalan's $G = L(\chi_{-4}, 2)$ for $\mathbf{E}$. Every entry of the limit column of the sporadic table is the character data of its source, read off.

We state this as an explanation, not a theorem: the precise Eichler–Shimura evaluation at the fold, including the rational prefactor (1/6, 7/32, 5/8, ...) and the second-kind (quasiperiod) contribution, is exactly the calculation that the programme's cuspidal companion work [4] shows is subtler than it looks — there, the fold value is a period-polynomial part *plus* a quasiperiod times a second-kind part, and for Eisenstein sources the second part is elementary and cancels. Writing the general law is open.

6.3. **The reverse factory has a sharp target.** Sol's "reverse factory" asks: can one *build* recurrences whose limits are new, interesting constants — L-values of elliptic curves, in particular? Verification 3.1 answers a preliminary question precisely: **not from the sporadic table**. A cuspidal source is required for an elliptic L-value, and the sporadic class, at q^{60}, provably-at-that-range contains no cuspidal component whatsoever.

The construction direction is therefore forced: choose a level with $S_{r+1} \neq 0$, pick a cusp form f there, set $B := F\theta_q^{-r} f$, and derive the recurrence it satisfies (which will be inhomogeneous). The discriminating question then becomes the *integrality/denominator* behaviour of that object, not its limit. The programme's cuspidal-companion experiments [4] do exactly this on Apéry's own curve and on Domb's, and find that cuspidality costs nothing in denominators — but that the convergence rate (4^{-n}) is too slow for irrationality. The present ledger's contribution is negative but clarifying: the sporadic table will not answer the denominator question, because it contains no cuspidal example to learn from.

6.4. **A denominator program.** The explicit $c(m)$ laws make one more thing attackable that was not before. Since $B_n = [t^n]F\Theta$ with $\Theta = \sum c(m) m^{-r} q^m$, the p-adic valuation $v_p(\operatorname{den} B_n)$ should be readable from the p-parts of $c(m)/m^r$ — the Bernoulli/Eisenstein constants of the source. This is Sol's step 4 of the flagship in the denominator direction. [OPEN]; not attempted.

7. The gap between q^{60} and a theorem

The programme never presents a finite computation as a proof, so it owes a precise account of what a proof would need. For Verification 3.1 it is short, and worth stating exactly, because it is a bookkeeping task rather than a mathematical obstacle.

7.1. **Sturm bounds.** If two forms in $M_k(\Gamma_0(N), \chi)$ agree in q-expansion up to the Sturm bound $\frac{k}{12}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$, they are equal. The bounds for the twelve identifications are *recorded* as follows.

fam	$\mathbf{A}$	$\mathbf{B}$	$\mathbf{C}$	$\mathbf{D}$	$\mathbf{E}$	$\mathbf{F}$	α	γ	δ	ε	ζ	η
weight	3	3	3	3	3	3	4	4	4	4	4	4
level	6	36	6	5	8	12	12	6	12	8	9	20
Sturm bound	3	18	3	1.5	3	6	8	4	8	4	4	12

Every bound is ≤ 18, against a verification to q^{60} — a margin of more than a factor of three even in the worst case ($\mathbf{B}$, level 36).

7.2. **What is therefore missing.** Each identification in Table 1 is a theorem *conditional on*:

(i) the certification that the eta / generalized-eta expressions of [2] for t and F are modular of the stated weight, level and character, and that $F = y_0(t)$ (i.e. that both sides satisfy the family's ODE); and
(ii) the certification that the stated divisor-sum combination lies in the corresponding Eisenstein space — classical, via the standard basis of $E_k^{\chi,\psi}(q^\ell)$.

Item (ii) is textbook. Item (i) is the standard eta-quotient order-at-cusps bookkeeping, and it is **done only for ζ (level 9)** so far; the other eleven are routine and undone. That is the entire gap.

Status of Table 1: [VERIFIED: q^{60}] for twelve families; [THM] for none. Only the ζ family's parametrization has its eta certification (i) in hand, and even there the identification is promoted only when (ii) is written out. The claim "the sporadic sources are Eisenstein" is a verification, not a theorem.

Promotion is a per-family finite bookkeeping campaign, requiring no new mathematics. It is the natural next piece of work.

7.3. **And the rest.** Lemma 4.1 is proved here. Verifications 4.4 and 5.2 are floating-point with tracked truncation, not exact; they confirm the mechanism and, in the four sharp rows of Table 2, confirm it to 8–15 digits. Verification 5.3 is a bounded search, so it excludes relations within its basis and coefficient cap and *nothing more* — "apparently new constants" means exactly that. Verification 3.2 excludes fits in the stated bases only, and specifically does *not* say Cooper's three are non-Eisenstein.

8. Evidence status: every claim of this paper

Claim	Label
$B = F\theta_q^{-r}\Phi$, $\Phi = t\sigma^r/(PF)$, all fifteen families, all n (Thm 1.1, *proved in* [1], not here)	[THM]
Fifteen nomes integral at $\mu = 1$; twelve (t, F) pairs are eta-type on the stated levels (Ver. 2.1, from [2])	[VERIFIED: q^{25}]
The twelve coefficient laws $c(m)$ of Table 1	[VERIFIED: q^{60}, held-out q^{27}–q^{60}]
No cuspidal component in any of the twelve sources	[VERIFIED: q^{60}, stated cusp bases]
Constant terms cancel identically in each Eisenstein combination	[PROVED] (forced by $\Phi = q + O(q^2)$)
γ reproduces Beukers' level-6 weight-4 series; $\mathbf{D}$ reproduces the $\Gamma_1(5)$ weight-3 picture	[VERIFIED: q^{60}] (controls)
ζ's source is the primitive (χ_{-3}, χ_{-3}) weight-4 level-9 Eisenstein series with coefficient 1	[VERIFIED: q^{60}]
No weight-3 or weight-4 fit for s_7, s_{10}, s_{18} in the levels $\{7, 10, 18\}$ and divisors, natural characters	[EXCLUDED: stated bases]
Φ integral ($\mu = 1$) for Cooper's three; the obstruction is the coordinate, not integrality	[VERIFIED: q^{26}] + [OPEN]
Identification of $\Phi_{s_7}, \Phi_{s_{10}}, \Phi_{s_{18}}$ (Prob. 3.3)	[OPEN]
Fold connection lemma $\xi = \Theta(q_c) + F\Theta'/F'$ (Lem. 4.1)	[PROVED]
Nine known Apéry limits recovered from the sources (Table 2; γ to $6.6 \cdot 10^{-15}$)	[VERIFIED: numeric, $N = 26$, tracked]
$\mathbf{B}, \delta, \eta$ are exactly the complex-root families, disc $-27, -128, -16$ (Obs. 5.1)	[PROVED]
Complex Eichler values $\xi_{\mathbf{B}}, \xi_\delta, \xi_\eta$ to 15/39/33 digits (Ver. 5.2)	[VERIFIED: numeric, $N = 60$, dps 80]
No rational relation for those values over the stated L-value bases (Ver. 5.3)	[EXCLUDED: basis, maxcoeff $10^{6\text{–}7}$]
Closed form for ξ_δ (Prob. 5.4)	[OPEN]
Sturm bounds ≤ 18 for the twelve spaces (§7)	recorded, [PROVED] (standard bound)
Eta certification (i) of §7 for the eleven non-ζ families	[OPEN] (routine, undone)
"The sporadic limits are ζ/Dirichlet L-values *because* the sources are Eisenstein" (§6)	explanation; general law [OPEN]
$v_p(\operatorname{den} B_n)$ readable from p-parts of $c(m)/m^r$	[OPEN], not attempted

Nothing above is claimed beyond its label. In particular the headline of this paper — "twelve of the fifteen sporadic companion sources are pure Eisenstein series" — is a [VERIFIED: q^{60}] statement and is not a theorem, and no sentence in this paper should be read as saying otherwise.

Acknowledgements

The programme of identifying the companion source as a modular object and deriving the Apéry limits from it was proposed by the collaborator Sol.

References

[1] [author], *Modular anchors for Apéry-like companions: second solutions as Eichler integrals, with the symmetric-square theorem for the sporadic operators*. Companion paper of this programme, `papers_out/modular_anchors`.

[2] `work/SPORADIC_MODULAR_DICTIONARY.md` — the fifteen pairs, their (t, F) parametrizations, and the ASD sweep. Scripts `work/z5eps/eps51_dictionary.py`, `eps51_refine.py`.

[3] `work/PHI_SOURCE_LEDGER.md` — the companion sources $\Phi(q)$ of the fifteen sporadic pairs. Scripts `work/z5eps/eps60_phi_source.py`, `eps60b_phi_holdouts.py`, `eps61_eichler_limits.py`, `eps61b_complex_limits.py`.

[4] `work/CUSPIDAL_COMPANION.md` — the first deliberately cuspidal companion, Sturm arithmetic for the twelve identifications, and the fold lemma as originally recorded. Scripts `work/z5eps/eps62*`.

[5] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$*, Astérisque **61** (1979), 11–13.

[6] F. Beukers, *Irrationality proofs using modular forms*, Astérisque **147–148** (1987), 271–283.

[7] D. Zagier, *Integral solutions of Apéry-like recurrence equations*, in: Groups and Symmetries, CRM Proc. Lecture Notes **47** (2009), 349–366.

[8] S. Cooper, *Sporadic sequences, modular forms and new series for* $1/\pi$, Ramanujan J. **29** (2012), 163–183.

[9] G. Almkvist and W. Zudilin, *Differential equations, mirror maps and zeta values*, in: Mirror Symmetry V, AMS/IP Stud. Adv. Math. **38** (2006), 481–515.